2014 F=ma Exam Review

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Problem 4

The N_2 and N_3 ones make sense because exploding must lie in a line for 2 particles and a plane for 3 particles. It's really a question about how many particles for N_1. Turns out only 2, because you have 2 equations, momentum and energy. To have a solution means there can be at most 2 unknowns, v_{1f} and v_{2f}, which you need for the final KE of each particle.

Problem 6

$\sum F = 0$ means v_{cm} is constant, so use v_{cm}, not the v of the block. Find v_{cm} first:

$$v_{cm} = \frac{m_{\text{block}} v_{\text{block}}}{m_{\text{block}} + m_{\text{box}}} = \frac{2(5)}{2+10} = \frac{5}{6}, \qquad x = v_{cm} t = \frac{5}{6} \cdot 60 = 50.$$

Problem 14

We need to divide our expressions for $K_{\text{translational}}$ and K_{total}. These involve v and ω, which we can express with $v = at$ and $\omega = \alpha t$. (That's safe because they do share time, so it'll cancel, whereas they don't share displacement, so don't use that: $\Delta x \neq r\Delta\theta$.) $a = F/m$ and $\alpha = FR/I$. Plug those in, plug v and ω in, and simplify the expression.

Problem 15

The question is just worded complicated. It gives τ, ω, R, and μ, but none of that is really used; they're distractors. μ may have been useful for frictional losses, but it's no slip. ω and R may have been useful if converting to v, which it appears to be, but start with $P = Fv$ and recognize that converting rotation to translation is not necessary here. F up the incline balances $mg\sin\theta$, $P = Fv$, so

$$v = \frac{P}{F} = \frac{P}{mg\sin\theta} = \frac{2P}{mg} \qquad (\theta = 30^\circ \text{ here}).$$

Problem 19

The friction/drag force is proportional to $\rho A v^2$, but ρ and v^2 are constant here, so just the cross-sectional area, which is proportional to length. $F_g = \lambda L g$, also proportional to length. The tension's vertical and horizontal components need to balance out friction and gravity. $\tan\theta = T_x/T_y$, and T_x and T_y are both proportional to length, so that cancels out: $\tan\theta = f/\lambda g$, so the angle is constant.

Problem 21

Write out the force and torque equations for the pulley and masses. Solve for a (recall $\alpha = a/R$ for a no-slip pulley) so you can divide the expressions to get the ratio (the goal in the question). Once you have the expression for a, you need expressions for I because that's the one thing that changes. For a disk with a circle removed from the center, just subtract the I that a disk of that shape would have had.

Problem 22

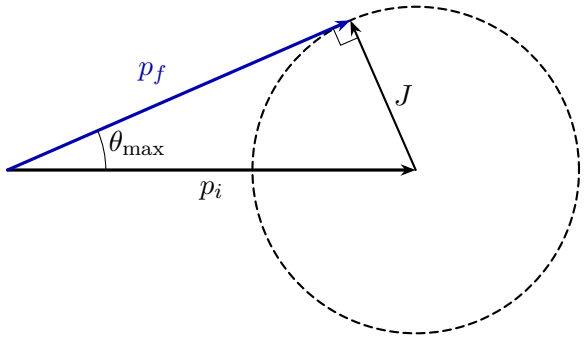
Kepler's third law for orbiting the com ($M + m$ in the denominator).

Problem 23

It gives the relative speed of the ball moving away, because they both have a certain velocity in the lab frame. $J = m\Delta v$, so find the change in the astronaut's velocity.

Problem 24

A fixed J can be fired in any direction, forming a circle of possible directions. The angle of the final v with the initial is the same as the angle of the final p to the initial, which is maximized when the final p is tangent to that circle. A diagram is useful here (they even hint at it).



Problem 25

Read the question carefully: μ is between the block and the second ramp, so friction only applies to one. Set up energy and geometry and solve for h_2.